APPENDIX

VII. DETAILED NUMERICAL RESULTS OF NOWCASTING EFFECTIVENESS

Tables IV and V report the detailed numerical values used for Figures 2 and 3 in the main paper, respectively. As mentioned in Section V-B, MULTIAIR outperforms all baseline methods across all lead times, in terms of both classification and regression metrics.

VIII. DETAILED NUMERICAL RESULTS OF ABLATION STUDY

Tables VII and VIII report the detailed numerical results of the ablation study in terms of classification and regression metrics, respectively. As discussed in Section V-C, MULTIAIR achieves the best overall performance, showing the effectiveness of its key components.

IX. HANDLING OF MISSING VALUES

In the case of time series data, missing values at each station $s \in S_T \cup S_C$ at $t' \in \{t - T_t, ..., t\}$ are filled with valid observations from the nearest station at t'. In this case, to distinguish between actual observed values and imputed values, a missing value indicator $\boldsymbol{m}_s^{(t')} \in \{0, 1\}^{d_t}$ is introduced, where d_t denotes the number of observed values in a ground observation sensor. This multi-hot vector indicates which values among $PM_{2.5}$ value and other observation values are originally missing for each station s at time t' and is used in constructing the input time series data. For AOD images, missing values do not occur independently like ground observations but can appear simultaneously over a wide area, making the proposed imputation method inaccurate. Thus, we fill the missing values with -1, as it explicitly falls outside the typical AOD range (about 0 to 4).

X. ADDITIONAL EXPERIMENTAL SETTINGS

In this section, we introduce detailed settings for training MULTIAIR and baseline methods in Section V-A, including hyperparameters.

For pre-training image models DeepRaNE and SimVP, we followed the training scheme of ASOC+ [16]. We used Mean Squared Error (MSE) loss and the Adam [31] optimizer with a learning rate 2×10^{-5}. We set the batch size to 20 and the number of training steps to 50,000. For every 1,000 steps, we evaluated the model by calculating loss on the validation set and selected the model with the best performance. The number of hidden channels of DeepRaNE was set to 32.

We used the modified focal-R L1 loss in Section IV-D with hyperparameters $\beta = 0.25$ and $\gamma = 2$ and used the Adam optimizer with a learning rate 10^{-3} for all other training processes. We set the batch size to 20 for ASOC+ and 120 for other models. We set the maximum number of epochs to 200 with early stopping if there was no performance gain for 5 epochs. For ASOC+, we set the hidden dimension for LSTM cells to 64 and the number of heads for the attention blocks to 4. For MULTIAIR, we set the hidden dimension for LSTM cells to 128.

We reproduced the architectures of CrossViViT and GNN-LSTM from [15] and [14], respectively, and adapted their implementations to match our experimental settings. Our task requires station-wise multivariate forecasting over multiple monitoring stations; therefore, we adopt CrossViViT-MS, a station-wise multi-station adaptation of CrossViViT, as a multimodal baseline. For GNN-LSTM, we constructed graphs with stations as nodes, edges between stations, and the speed and direction of wind as the edge weights. To ensure a fair comparison with our method, we used only the ground-based observations on air quality and AOD image.

For all time-series forecasting baseline methods, we follow their original hyperparameter settings without additional tuning. Specifically, the Transformer uses a model dimension of 64 with 8 attention heads and 2 layers. TimesNet uses 2 TimesBlocks with a feed-forward dimension of 32, while TimeMixer uses 2 PDM blocks with a feed-forward dimension of 32.

XI. EVALUATION USING EXTRA METRICS

In this section, we introduce evaluation results using extra evaluation metrics. For extra evaluation metrics, we use four classification metrics: accuracy (ACC), probability of detection (POD), false alarm rate (FAR), and F1 score, and three regression metrics: RMSE, normalized mean error (NME), and Pearson correlation coefficient (ρ). For extra evaluation metrics, we simply evaluate once for all prediction lead times. ACC is defined as the ratio of correct class prediction based on 4 classes in Section V-A: *Severity 1, 2, 3* and *4*. Other classification metrics, POD, FAR, and F1 score, focus on predicting extreme air pollution of *Severity 3 or worse*, which are defined as follows:

$$\text{POD} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{FAR} = \frac{\text{FP}}{\text{TP} + \text{FP}},$$
$$\text{F1} = \frac{2 \times \text{TP}}{2 \times \text{TP} + \text{FP} + \text{FN}},$$

where TP, FP, FN stand for the counts of true positives, false positives, and false negatives, respectively, for air pollution of *Severity 3 or worse*. NME, ρ is defined as:

$$\text{NME} = \frac{\sum |\hat{\mathbf{P}}_T - \mathbf{P}_T|}{\sum \mathbf{P}_T} \times 100,$$
$$\rho = \frac{\sum((\hat{\mathbf{P}}_T - \mu_{\hat{\mathbf{P}}_T}) \times (\mathbf{P}_T - \mu_{\mathbf{P}_T}))}{\sqrt{\sum(\hat{\mathbf{P}}_T - \mu_{\hat{\mathbf{P}}_T})^2 \times \sum(\mathbf{P}_T - \mu_{\mathbf{P}_T})^2}},$$

where $\mathbf{P}_T, \hat{\mathbf{P}}_T$ each denotes all ground truth $PM_{2.5}$ values and predicted $PM_{2.5}$ values respectively, and $\mu_{\mathbf{P}_T}$, $\mu_{\hat{\mathbf{P}}_T}$ each denotes the mean value of ground truth $PM_{2.5}$ values and predicted $PM_{2.5}$ values respectively.

Table IV, Table V and Table VI present the evaluation results in comparison with the baseline models. MULTIAIR performs the best for every metric except for FAR. However, AirFormer, which achieves the best FAR, records the lowest POD and F1 score among the models, suggesting that it simply underestimates severe air pollution.

TABLE IV

PREDICTIVE PERFORMANCES ON THE TEST SET EVALUATED USING CLASSIFICATION METRICS (CSI AND F1 SCORES, WHERE HIGHER VALUES INDICATE BETTER PERFORMANCE). MULTIAIR PERFORMS BEST ACROSS ALL POLLUTION LEVELS AND LEAD TIMES. IN EACH SETTING, THE BEST AND THE SECOND-BEST PERFORMANCES ARE IN **BOLD** AND UNDERLINED.

Pollution level	Lead time	**MULTIAIR** CSI	**MULTIAIR** F1	ASOC+ CSI	ASOC+ F1	CrossViViT-MS CSI	CrossViViT-MS F1	LSTM CSI	LSTM F1	Transformer CSI	Transformer F1	TimesNet CSI	TimesNet F1	TimeMixer CSI	TimeMixer F1	AirFormer CSI	AirFormer F1	GNN-LSTM CSI	GNN-LSTM F1
Severity 2 or worse ($> 15\mu g/m^3$)	1 hr.	**0.822**	**0.902**	0.720	0.837	0.806	0.892	0.804	0.891	0.793	0.885	0.697	0.822	0.765	0.867	0.755	0.861	0.815	0.898
	2 hr.	**0.786**	**0.880**	0.707	0.828	0.759	0.863	0.755	0.861	0.752	0.859	0.687	0.815	0.735	0.848	0.694	0.819	0.767	0.868
	3 hr.	**0.757**	**0.862**	0.689	0.816	0.723	0.839	0.718	0.836	0.718	0.836	0.677	0.807	0.711	0.831	0.646	0.785	0.727	0.842
	4 hr.	**0.732**	**0.845**	0.670	0.802	0.695	0.820	0.690	0.816	0.688	0.815	0.664	0.798	0.688	0.815	0.596	0.747	0.692	0.818
	5 hr.	**0.708**	**0.829**	0.651	0.788	0.672	0.804	0.664	0.798	0.663	0.797	0.652	0.798	0.667	0.800	0.557	0.716	0.660	0.795
	6 hr.	**0.688**	**0.815**	0.631	0.774	0.653	0.791	0.644	0.784	0.641	0.782	0.640	0.781	0.651	0.788	0.521	0.685	0.632	0.775
Severity 3 or worse ($> 35\mu g/m^3$)	1 hr.	**0.756**	**0.861**	0.567	0.724	0.696	0.821	0.726	0.842	0.680	0.810	0.562	0.720	0.646	0.785	0.654	0.791	0.725	0.840
	2 hr.	**0.686**	**0.814**	0.542	0.703	0.596	0.747	0.628	0.772	0.614	0.761	0.543	0.704	0.594	0.745	0.569	0.725	0.618	0.764
	3 hr.	**0.633**	**0.775**	0.514	0.679	0.512	0.677	0.554	0.713	0.558	0.717	0.519	0.683	0.551	0.711	0.498	0.665	0.543	0.704
	4 hr.	**0.589**	**0.741**	0.487	0.655	0.452	0.623	0.498	0.665	0.511	0.677	0.490	0.658	0.513	0.678	0.443	0.614	0.487	0.655
	5 hr.	**0.552**	**0.711**	0.459	0.629	0.404	0.575	0.454	0.624	0.472	0.641	0.466	0.636	0.483	0.651	0.392	0.563	0.441	0.612
	6 hr.	**0.519**	**0.683**	0.431	0.603	0.368	0.538	0.419	0.590	0.438	0.610	0.446	0.617	0.459	0.629	0.338	0.505	0.402	0.573
Severity 4 ($> 75\mu g/m^3$)	1 hr.	**0.699**	**0.823**	0.322	0.487	0.593	0.744	0.648	0.787	0.561	0.719	0.305	0.467	0.537	0.699	0.638	0.779	0.597	0.748
	2 hr.	**0.601**	**0.751**	0.287	0.446	0.468	0.637	0.505	0.671	0.487	0.655	0.273	0.429	0.465	0.634	0.501	0.668	0.459	0.629
	3 hr.	**0.527**	**0.691**	0.251	0.401	0.361	0.530	0.405	0.576	0.424	0.595	0.226	0.369	0.405	0.576	0.397	0.568	0.356	0.526
	4 hr.	**0.471**	**0.641**	0.217	0.357	0.282	0.440	0.339	0.506	0.372	0.542	0.187	0.316	0.344	0.512	0.314	0.478	0.281	0.439
	5 hr.	**0.418**	**0.590**	0.186	0.314	0.225	0.367	0.287	0.446	0.327	0.493	0.158	0.273	0.306	0.468	0.240	0.388	0.224	0.367
	6 hr.	**0.369**	**0.539**	0.157	0.217	0.177	0.301	0.232	0.377	0.289	0.449	0.144	0.251	0.276	0.433	0.176	0.300	0.176	0.299
Average		**0.629**	**0.764**	0.472	0.614	0.524	0.667	0.554	0.698	0.555	0.702	0.463	0.608	0.544	0.693	0.496	0.648	0.533	0.675

TABLE V

PREDICTIVE PERFORMANCES ON THE TEST SET EVALUATED USING REGRESSION METRICS (RMSE AND MAE, WHERE LOWER VALUES INDICATE BETTER PERFORMANCE). MULTIAIR PERFORMS BEST ACROSS ALL LEAD TIMES. IN EACH SETTING, THE BEST AND THE SECOND-BEST PERFORMANCES ARE IN **BOLD** AND UNDERLINED.

Lead time	**MULTIAIR** RMSE	**MULTIAIR** MAE	ASOC+ RMSE	ASOC+ MAE	CrossViViT-MS RMSE	CrossViViT-MS MAE	LSTM RMSE	LSTM MAE	Transformer RMSE	Transformer MAE	TimesNet RMSE	TimesNet MAE	TimeMixer RMSE	TimeMixer MAE	AirFormer RMSE	AirFormer MAE	GNN-LSTM RMSE	GNN-LSTM MAE
1 hr.	**3.998**	**2.727**	7.180	4.734	4.925	3.244	4.539	3.051	5.182	3.427	7.431	5.139	5.821	3.871	5.692	4.277	7.723	4.922
2 hr.	**5.184**	**3.440**	7.601	4.990	6.547	4.194	6.146	4.010	6.407	4.161	7.817	5.389	6.842	4.509	7.013	5.026	10.247	6.696
3 hr.	**6.108**	**3.992**	8.111	5.298	7.872	4.948	7.308	4.727	7.455	4.801	8.249	5.644	7.697	5.029	8.132	5.694	12.204	8.084
4 hr.	**6.886**	**4.463**	8.624	5.609	8.865	5.537	8.184	5.280	8.366	5.373	8.750	5.944	8.483	5.536	9.100	6.306	13.774	9.207
5 hr.	**7.570**	**4.885**	9.120	5.914	9.659	6.014	8.902	5.735	9.175	5.889	9.206	6.226	9.161	5.971	9.889	6.779	15.146	10.187
6 hr.	**8.160**	**5.250**	9.580	6.206	10.255	6.402	9.481	6.112	9.874	6.340	9.572	6.485	9.648	6.288	10.665	7.264	16.367	11.067
Average	**6.318**	**4.126**	8.369	5.459	8.021	5.056	7.427	4.819	7.743	4.999	8.504	5.805	7.942	5.201	8.415	5.891	12.577	8.360

TABLE VI

PREDICTIVE PERFORMANCES ON THE TEST SET IN TERMS OF EXTRA METRICS. FOR METRICS WITH AN UPWARD ARROW (↑), HIGHER VALUES CORRESPOND TO BETTER PERFORMANCE, WHEREAS FOR THOSE WITH A DOWNWARD ARROW (↓), LOWER VALUES ARE PREFERABLE. IN EACH SETTING, THE BEST AND THE SECOND-BEST PERFORMANCES ARE IN **BOLD** AND UNDERLINED.

Metric		MULTIAIR	ASOC+	CrossViViT-MS	LSTM	Transformer	TimesNet	TimeMixer	AirFormer	GNN-LSTM
Class.	ACC (↑)	**0.812**	0.749	0.772	0.778	0.772	0.726	0.763	0.753	0.784
	POD (↑)	**0.728**	0.596	0.603	0.639	0.665	0.645	0.690	0.508	0.630
	FAR (↓)	0.192	0.248	0.211	0.212	0.256	0.304	0.289	**0.100**	0.224
	F1 (↑)	**0.766**	0.666	0.669	0.705	0.702	0.670	0.700	0.650	0.695
Reg.	RMSE (↓)	**6.464**	8.410	8.226	7.613	7.906	8.537	8.050	8.407	7.938
	NME (↓)	**22.146**	29.346	27.166	25.907	26.871	31.208	27.959	31.670	26.897
	ρ (↑)	**0.902**	0.825	0.834	0.859	0.847	0.817	0.845	0.874	0.846

Table VII, Table VIII and Table IX present the evaluation results in comparison with the ablation models, which are described in Section V-C. MULTIAIR achieves the best or second-best performance for most metrics, including ACC, FAR, F1, RMSE, NME, and ρ. Although several ablation variants yield slightly higher POD, they generally incur higher FAR and lower F1, indicating a less balanced trade-off between detecting severe pollution and avoiding false alarms.

TABLE VII

ABLATION STUDY RESULTS BASED ON CLASSIFICATION METRICS (CSI AND F1, WHERE HIGHER VALUES INDICATE BETTER PERFORMANCE). ON AVERAGE, MULTIAIR PERFORMS BEST IN BOTH METRICS. IN EACH SETTING, THE BEST AND SECOND-BEST PERFORMANCES ARE IN **BOLD** AND UNDERLINED.

Pollution level	Lead time	MULTIAIR CSI	MULTIAIR F1	w/o Sat. CSI	w/o Sat. F1	w/o Dec. CSI	w/o Dec. F1	w/o LN CSI	w/o LN F1	w/o Curr. CSI	w/o Curr. F1	w/o HP CSI	w/o HP F1	w/o HP+Under CSI	w/o HP+Under F1	w/o HP+Over CSI	w/o HP+Over F1
Severity 2 or worse ($> 15\mu g/m^3$)	1 hr.	**0.822**	**0.902**	0.821	0.901	0.819	0.900	0.821	**0.902**	0.816	0.899	0.817	0.899	0.817	0.900	0.817	0.900
	2 hr.	0.786	0.880	0.784	0.879	0.780	0.876	**0.787**	**0.881**	0.780	0.876	0.782	0.878	0.783	0.878	0.784	0.879
	3 hr.	0.757	0.862	0.755	0.861	0.749	0.857	**0.759**	**0.863**	0.752	0.858	0.754	0.859	0.756	0.861	0.756	0.861
	4 hr.	0.732	0.845	0.731	0.844	0.723	0.839	**0.736**	**0.848**	0.728	0.842	0.728	0.843	0.733	0.846	0.733	0.846
	5 hr.	0.708	0.829	0.709	0.830	0.698	0.822	**0.714**	**0.833**	0.706	0.827	0.705	0.827	0.713	**0.833**	0.712	0.832
	6 hr.	0.688	0.815	0.690	0.816	0.677	0.807	0.694	**0.820**	0.686	0.814	0.684	0.813	**0.695**	**0.820**	0.694	0.819
Severity 3 or worse ($> 35\mu g/m^3$)	1 hr.	**0.756**	**0.861**	0.753	0.859	0.748	0.856	0.754	0.860	0.747	0.855	0.749	0.857	0.748	0.856	0.745	0.854
	2 hr.	**0.686**	**0.814**	0.680	0.810	0.671	0.803	0.685	0.813	0.673	0.805	0.677	0.807	0.675	0.806	0.672	0.804
	3 hr.	**0.633**	**0.775**	0.625	0.769	0.618	0.764	0.631	0.773	0.621	0.766	0.625	0.769	0.622	0.767	0.618	0.764
	4 hr.	**0.589**	**0.741**	0.581	0.735	0.575	0.730	0.584	0.737	0.578	0.732	0.582	0.736	0.579	0.733	0.573	0.729
	5 hr.	**0.552**	**0.711**	0.542	0.703	0.536	0.698	0.544	0.705	0.540	0.701	0.544	0.705	0.542	0.703	0.535	0.697
	6 hr.	**0.519**	**0.683**	0.509	0.675	0.504	0.670	0.509	0.675	0.509	0.674	0.510	0.675	0.511	0.676	0.501	0.668
Severity 4 ($> 75\mu g/m^3$)	1 hr.	**0.699**	**0.823**	0.692	0.818	0.683	0.812	0.682	0.811	0.685	0.813	0.666	0.800	0.667	0.800	0.659	0.795
	2 hr.	**0.601**	**0.751**	0.590	0.742	0.569	0.726	0.568	0.725	0.579	0.733	0.559	0.717	0.548	0.708	0.538	0.700
	3 hr.	**0.527**	**0.691**	0.507	0.673	0.493	0.660	0.475	0.644	0.501	0.667	0.479	0.647	0.461	0.631	0.454	0.625
	4 hr.	**0.471**	**0.641**	0.441	0.612	0.434	0.606	0.399	0.571	0.439	0.610	0.418	0.589	0.388	0.560	0.386	0.558
	5 hr.	**0.418**	**0.590**	0.380	0.551	0.381	0.552	0.344	0.512	0.384	0.554	0.357	0.526	0.332	0.498	0.332	0.499
	6 hr.	**0.369**	**0.539**	0.327	0.493	0.333	0.500	0.296	0.457	0.334	0.500	0.300	0.461	0.278	0.435	0.280	0.438
Average		**0.629**	**0.764**	0.618	0.754	0.611	0.749	0.610	0.746	0.614	0.751	0.608	0.745	0.603	0.740	0.599	0.737

TABLE VIII

ABLATION STUDY RESULTS BASED ON REGRESSION METRICS (RMSE AND MAE, WHERE LOWER VALUES INDICATE BETTER PERFORMANCE). IN EACH SETTING, THE BEST AND THE SECOND-BEST PERFORMANCES ARE IN **BOLD** AND UNDERLINED.

Lead time	MULTIAIR RMSE	MULTIAIR MAE	w/o Sat. RMSE	w/o Sat. MAE	w/o Dec. RMSE	w/o Dec. MAE	w/o LN RMSE	w/o LN MAE	w/o Curr. RMSE	w/o Curr. MAE	w/o HP RMSE	w/o HP MAE	w/o HP+Under RMSE	w/o HP+Under MAE	w/o HP+Over RMSE	w/o HP+Over MAE
1 hr.	**3.998**	**2.727**	4.038	2.753	4.117	2.791	4.051	2.757	4.154	2.825	4.149	2.830	4.162	2.841	4.167	2.843
2 hr.	**5.184**	**3.440**	5.253	3.482	5.401	3.573	5.207	3.449	5.390	3.571	5.359	3.556	5.394	3.562	5.403	3.564
3 hr.	**6.108**	3.992	6.198	4.045	6.372	4.163	6.138	**3.980**	6.345	4.145	6.295	4.120	6.340	4.117	6.316	4.110
4 hr.	**6.886**	4.463	7.003	4.521	7.197	4.678	6.949	**4.435**	7.172	4.642	7.063	4.596	7.121	4.581	7.054	4.566
5 hr.	**7.570**	4.885	7.721	4.948	7.957	5.154	7.626	**4.825**	7.905	5.087	7.723	5.021	7.799	4.991	7.705	4.973
6 hr.	**8.160**	5.250	8.338	5.321	8.624	5.572	8.201	**5.178**	8.525	5.472	8.313	5.406	8.435	5.374	8.296	5.346
Average	**6.318**	4.126	6.425	4.178	6.611	4.322	6.362	**4.104**	6.582	4.290	6.484	4.255	6.542	4.244	6.490	4.234

TABLE IX

ABLATION STUDY RESULTS IN TERMS OF EXTRA METRICS. FOR METRICS WITH AN UPWARD ARROW (↑), HIGHER VALUES CORRESPOND TO BETTER PERFORMANCE, WHEREAS FOR THOSE WITH A DOWNWARD ARROW (↓), LOWER VALUES ARE PREFERABLE. IN EACH SETTING, THE BEST AND THE SECOND-BEST PERFORMANCES ARE IN **BOLD** AND UNDERLINED.

Metric		MULTIAIR	w/o Sat.	w/o Dec.	w/o LN	w/o Curr.	w/o HP	w/o HP+Under	w/o HP+Over
Class.	ACC (↑)	**0.812**	0.808	0.800	**0.812**	0.803	0.803	0.804	0.805
	POD (↑)	0.728	0.724	0.730	0.714	**0.739**	0.729	0.737	0.715
	FAR (↓)	0.192	0.201	0.220	**0.181**	0.226	0.208	0.220	0.202
	F1 (↑)	**0.766**	0.760	0.754	0.763	0.756	0.759	0.758	0.754
Reg.	RMSE (↓)	**6.464**	6.579	6.784	6.506	6.739	6.635	6.698	6.637
	NME (↓)	22.146	22.428	23.233	**22.031**	23.036	22.872	22.817	22.760
	ρ (↑)	**0.902**	0.897	0.890	0.899	0.892	0.894	0.892	0.894

Overall, MULTIAIR provides strong and balanced performance for extreme air pollution nowcasting.

XII. HYPERPARAMETER SENSITIVITY

In this section, we evaluate the sensitivity of several hyperparameters in MULTIAIR, including the input time-series

TABLE X

HYPERPARAMETER SENSITIVITY TEST RESULTS IN TERMS OF EXTRA METRICS. FOR METRICS WITH AN UPWARD ARROW (↑), HIGHER VALUES CORRESPOND TO BETTER PERFORMANCE, WHEREAS FOR THOSE WITH A DOWNWARD ARROW (↓), LOWER VALUES ARE PREFERABLE. IN EACH SETTING, THE BEST PERFORMANCE IS IN **BOLD**.

Metric		Input Length (T_t)				Hidden Dimension (d_h)			Gamma (γ)		
		4	7	13	25	64	128	256	1	2	3
Class.	ACC (↑)	0.801	**0.812**	0.811	**0.812**	0.805	**0.812**	0.801	0.806	**0.812**	0.808
	POD (↑)	0.730	0.729	0.727	**0.737**	0.728	0.729	**0.736**	**0.748**	0.729	0.728
	FAR (↓)	0.219	**0.192**	0.197	0.194	0.207	**0.192**	0.228	0.220	**0.192**	0.205
	F1 (↑)	0.754	0.766	0.763	**0.770**	0.759	**0.766**	0.754	0.764	**0.766**	0.760
Reg.	RMSE (↓)	6.712	6.444	6.463	**6.346**	6.672	**6.444**	6.755	6.571	**6.444**	6.539
	NME (↓)	35.747	**34.225**	34.716	35.765	35.383	**34.225**	36.460	35.443	**34.225**	34.617
	ρ (↑)	0.893	0.901	0.901	**0.905**	0.894	**0.901**	0.891	0.897	**0.901**	0.898

length, the number of hidden dimensions, and the focal loss parameter γ. As shown in Table X, increasing the input time-series length generally improves performance, although the gains are marginal. Similarly, the results indicate that MULTI-AIR is relatively insensitive to the choice of hidden dimension size and the value of γ, exhibiting stable performance across a diverse range of hyperparameter settings.